

Heterogeneous Treatment Effects

CML Week 5

2026-09-29

Schedule

1. Design-based causal inference & Monte Carlo simulation
2. Covariate adjustment
3. Balancing weights
4. Double robustness, Double ML and TMLE
5. **Heterogeneous Treatment Effects**
6. HTE II: Neural Networks
7. Off-policy evaluation and optimization
8. Experimental design
9. Panel data and the new diff-in-diff
10. Partial Identification
11. Response adaptation
12. Interference



Thus far

- Every week so far has targeted **one number**: the **average treatment effect** $\tau = \mathbb{E}[Y(1) - Y(0)]$
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But what if treatment effects vary?

- *Medical treatments*: some patients benefit, others don't (or are harmed)
- *Policy interventions*: context matters for effectiveness
- *Marketing*: not all customers respond the same way

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Goal today: understand why estimating heterogeneous effects is **fundamentally harder than it seems**, and where the ideas for doing it better come from.

(Holland 1986)

The Fundamental Problem of Causal Inference

> For any individual unit i, we can never observe both $Y_i(1)$ and $Y_i(0)$ simultaneously.

- Each unit has two potential outcomes: $Y_i(1)$ and $Y_i(0)$
- We only observe: $Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0)$
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Why it's fundamental:

- Not a sample size problem (more data doesn't help)
- Not a measurement problem (we can't measure what doesn't exist in our world)
- It's a metaphysical constraint on what we can learn from data



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Implications:

- We can make causal claims about populations or subgroups
- We cannot make causal claims about specific individuals with certainty
- All individual-level causal predictions are inherently model-dependent



What can we learn about individuals?

What we cannot identify:

- Individual treatment effect: $\tau_i = Y_i(1) - Y_i(0)$
- Whether a specific person benefited from treatment

What we can identify (under assumptions):

- Average Treatment Effect: $\tau = E[Y(1) - Y(0)]$
- Conditional Average Treatment Effect: $\tau(x) = E[Y(1) - Y(0) \mid X = x]$

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CATE as a compromise:

$$\tau(x) = E[Y(1) - Y(0) \mid X = x]$$

- Not about individuals, but about groups defined by X
- As X becomes more detailed, groups become smaller
- But we still need multiple units with similar X to estimate $\tau(x)$
- **Bias-variance tradeoff**: more granular X means smaller groups, higher variance

The role of randomization

For the ATE, randomization is powerful:

- Ensures $A \perp (Y(1), Y(0))$
- Simple difference in means is unbiased: $E[\bar{Y}_{A=1} - \bar{Y}_{A=0}] = \tau$
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The curse of dimensionality:

- With 10 binary covariates: $2^{10} = 1024$ subgroups
- Even in a large RCT, many subgroups may have no treated or control units
- Must **borrow strength across subgroups** (i.e., model)

How can we do machine learning?

The standard supervised learning paradigm:

1. Observe many examples of (X_i, Y_i)
2. Learn function $f : X \to Y$ that minimizes prediction error
3. Validate on held-out test set
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Why this fails for HTE:

- **No observed target:** we never see $\tau_i = Y_i(1) - Y_i(0)$
- **No natural loss function:** can't compute $(\hat{\tau}(X_i) - \tau_i)^2$ to optimize
- **Validation is impossible:** can't check if predictions are good on a test set

The problem with naive approaches

Consider the naive "difference in predictions" approach (**T-learner**):

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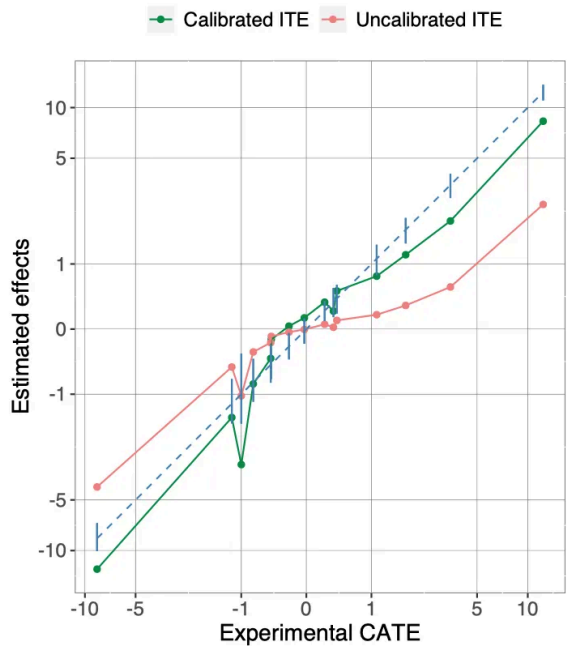
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The consequence: overfitting to noise in the heterogeneity, mistaking prognostic variables for moderators, and overconfident, miscalibrated predictions.
What we need: methods that explicitly target $\tau(x)$, not just Y

Bonus: (Leng and Dimmery 2024)

What can we observe about our HTE estimates?

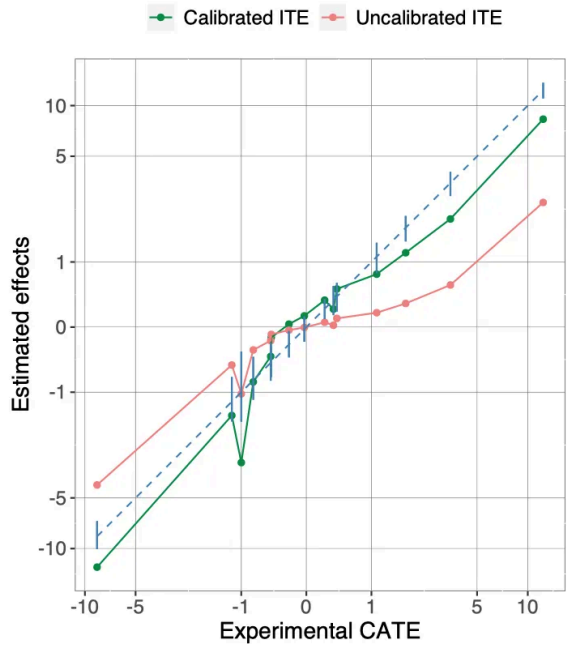
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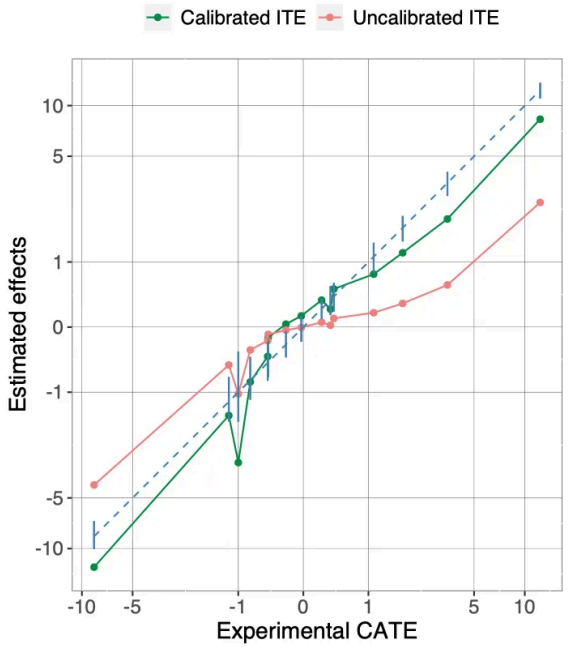
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Calibration check:

1. Sort units by $\hat{\tau}(X_i)$ into quantiles (e.g., deciles)
2. In each quantile q, compare the predicted effect $\bar{\hat{\tau}}_q = \frac{1}{n_q} \sum_{i \in q} \hat{\tau}(X_i)$ with the observed one $\bar{\tau}_q = \bar{Y}_{A=1,q} - \bar{Y}_{A=0,q}$
3. Plot predicted vs. observed

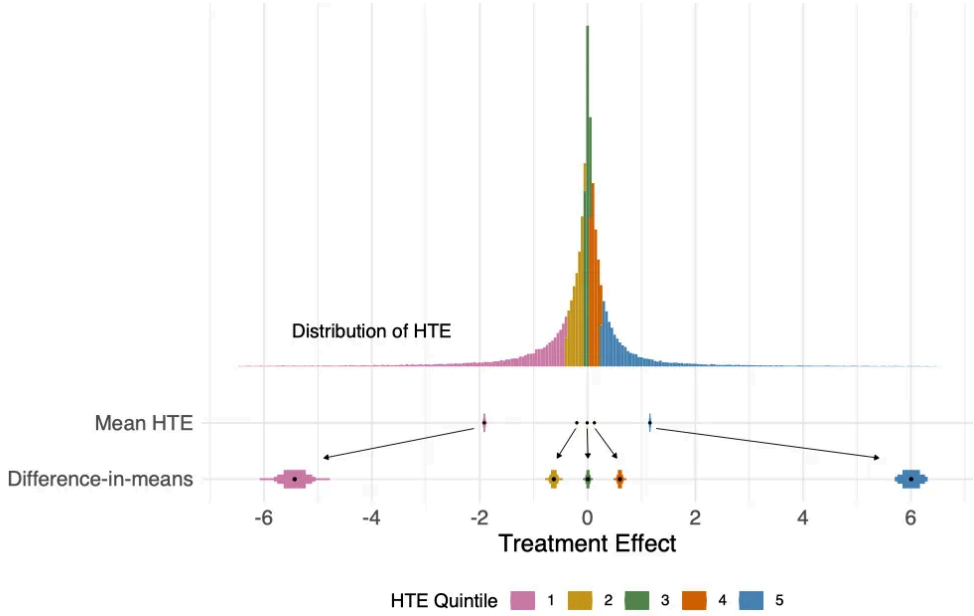
Perfect calibration: points lie on the 45-degree line



Calibration in practice

Evidence from large-scale experiments (Facebook, Criteo platforms):

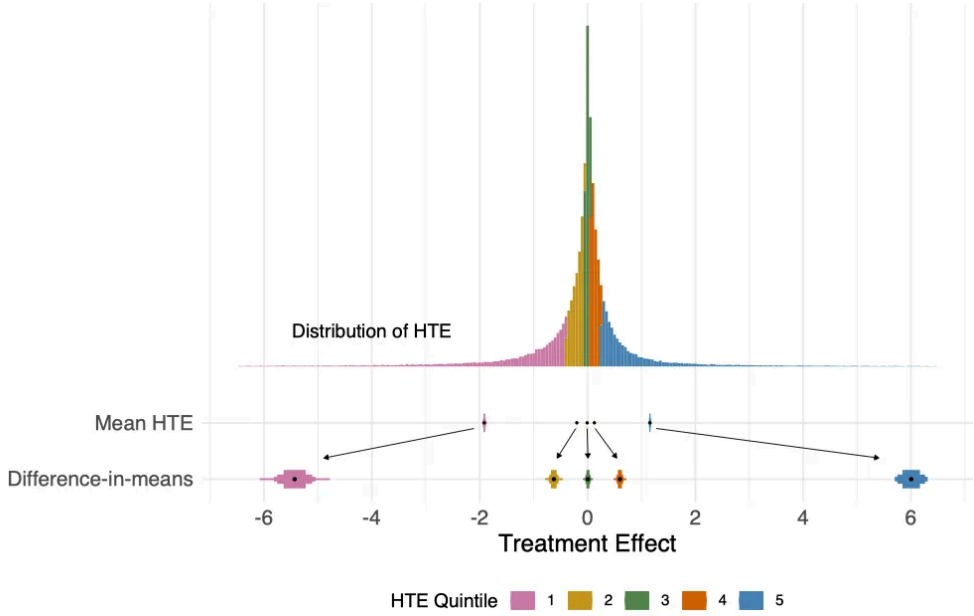
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Calibration in practice

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Why does this happen?

- Optimization targets prediction of Y, not τ
- Small treatment effects relative to outcome variance
- Prognostic/predictive confusion amplified by complex models

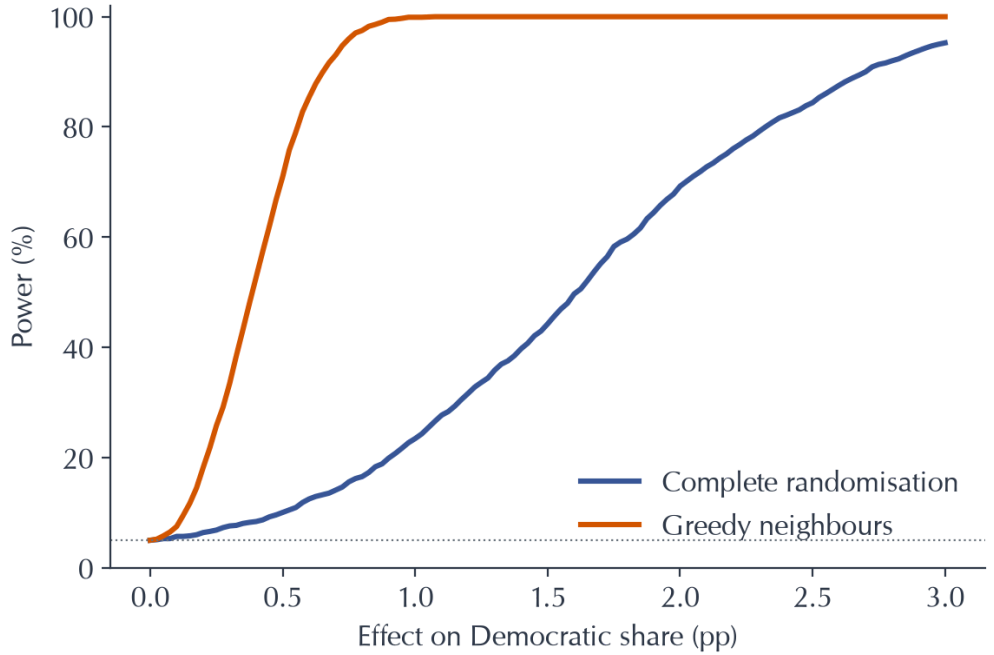
Bonus: (Arbour, Dimmery, and Rao 2021)

A complementary approach: can we do better if we *choose* how to assign treatment?

› Not all units are equally informative about $\tau(x)$

» Units with common covariate values: less valuable (we already have many like them)

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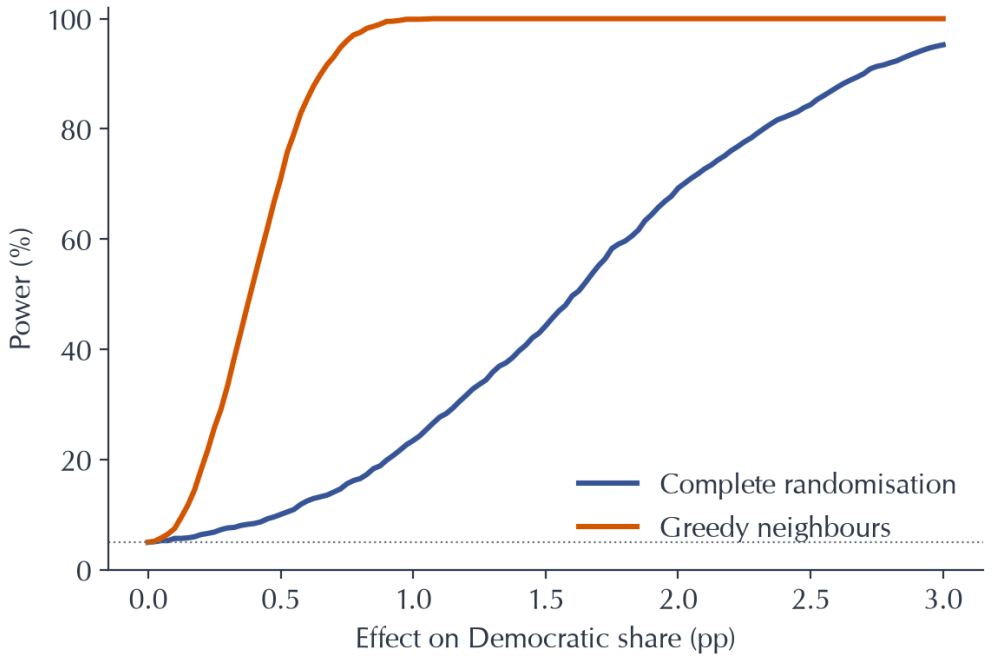
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Locally balanced assignments:

- Ensure good balance not just on average, but locally
- Assignments are well distributed across covariate space

Connection to today's themes:

- Design can't solve the Fundamental Problem
- But it can make sure you don't shoot yourself in the foot



Takeaways

- **The Fundamental Problem matters:**
  - We can never observe individual treatment effects
  - This makes HTE estimation fundamentally different from supervised learning
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- **What makes HTE hard:**
  - No ground truth for validation
  - Easy to overfit to noise
  - Prognostic vs. predictive confusion

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  - This makes HTE estimation fundamentally different from supervised learning
  - Standard ML intuitions can mislead us
- **What makes HTE hard:**
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  - Prognostic vs. predictive confusion
- **Practical implications:**
  - Need specialized methods (which students will present today)
  - Need calibration checks when possible
  - Skepticism is warranted for HTE claims



Open questions

1. **Adapting a workhorse:** Random forests predict Y well. Can a forest target $\tau(x)$ directly, and still deliver valid confidence intervals?
   - *Motivates:* (Wager and Athey 2018) on causal forests
2. **Combining models:** If we already have good outcome models, how should we combine them to get $\tau(x)$, and when does each recipe win?
   - *Motivates:* (Künzel et al. 2019) on meta-learners
3. **Double robustness for CATEs:** Last week's machinery hedged between two models for one number. Can it be pointed at $\tau(x)$, and how fast can we learn it?
   - *Motivates:* (Kennedy 2020) on the DR-learner
4. **Inference without a target:** If τ_i is never observed, what *can* we test about heterogeneity in an RCT with honest inference?
   - *Motivates:* (Chernozhukov et al. 2025) on generic ML inference

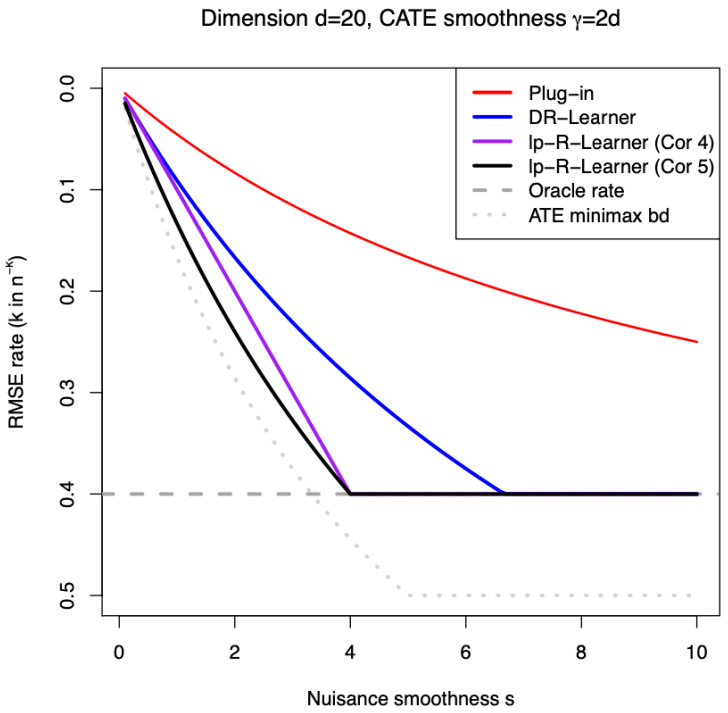
More to read

Paper	Why you'd read it
(Athey and Imbens 2019)	The review that frames today: the ML toolkit for applied economists, and where its causal uses part ways with prediction.
(Wager and Athey 2018)	Causal forests: *honest* trees (split on one half, estimate on the other) give consistency and asymptotic normality for $\tau(x)$.
(Künzel et al. 2019)	Meta-learners: S-, T- and X-learners as recipes for any base learner. The X-learner wins when the arms are unbalanced.
(Kennedy 2020)	The DR-learner: regress a doubly robust pseudo-outcome on X. The error is a product of nuisance errors, as last week.
(Chernozhukov et al. 2025)	Generic ML: test for heterogeneity (BLP), sort groups by it (GATES), describe them (CLAN), with valid inference after sample splitting.
(Athey and Imbens 2016); (Athey, Tibshirani, and Wager 2019)	The honest causal tree before causal forests, and the general framework (forests for any local moment condition) after.
(Nie and Wager 2021)	The R-learner: Robinson's residual-on-residual regression as a loss for $\tau(x)$, with quasi-oracle rates.

Deeper dive: (Kennedy 2020)

The DR-learner

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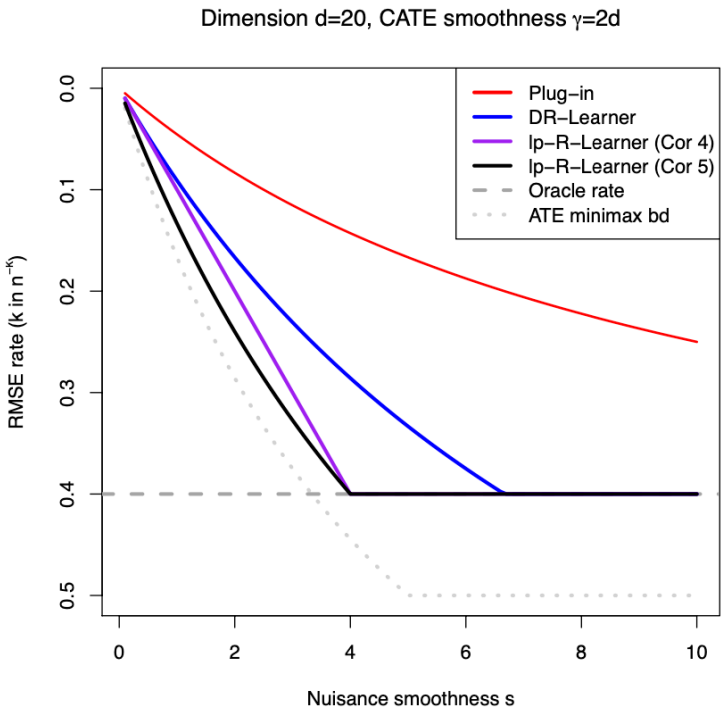


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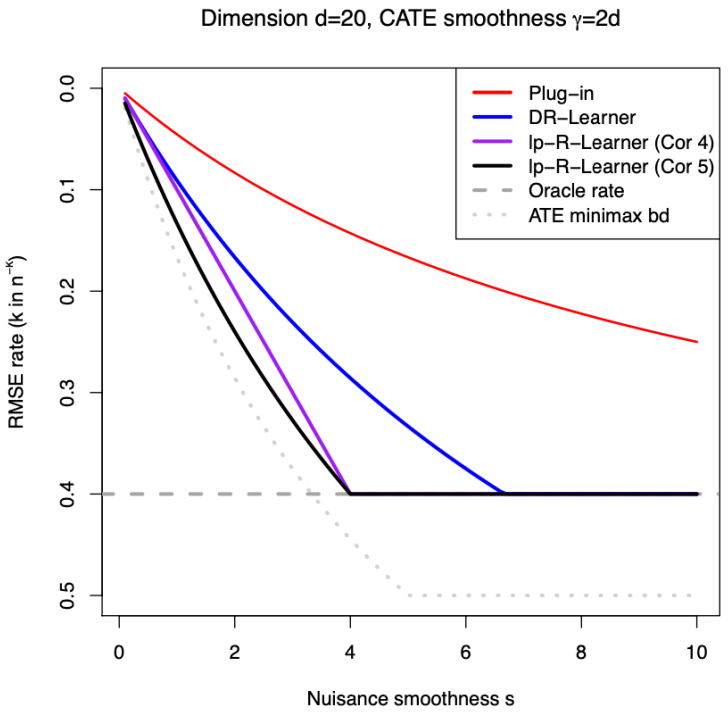
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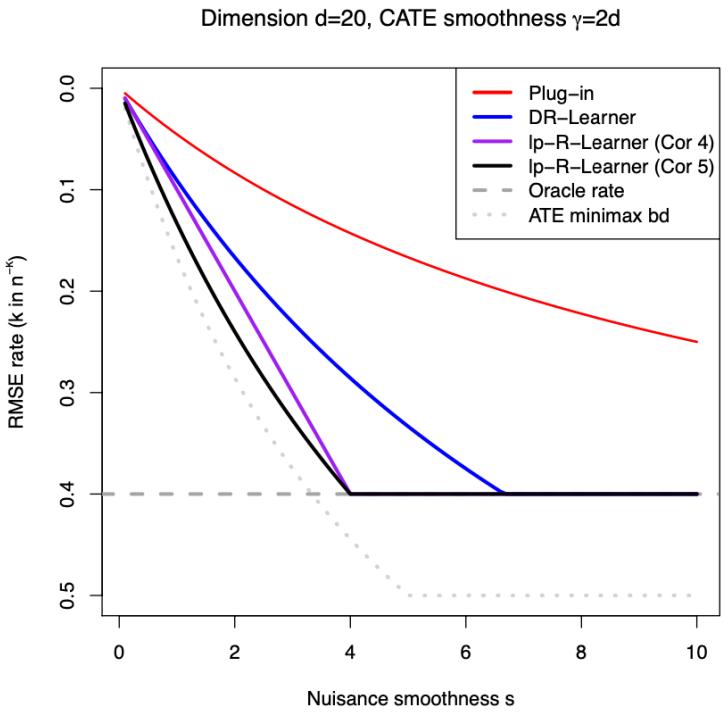
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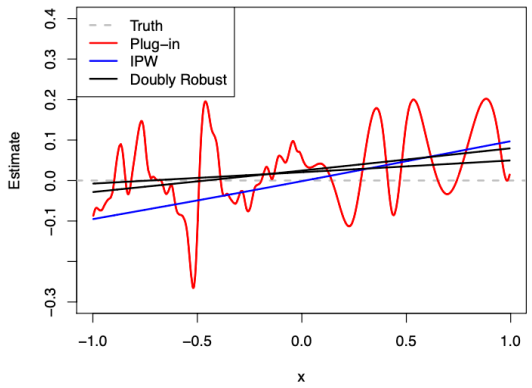
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› Nuisance training and smoothing must be on separate subsamples (**cross-fitting**); rotate the folds and average.



Why does this work?

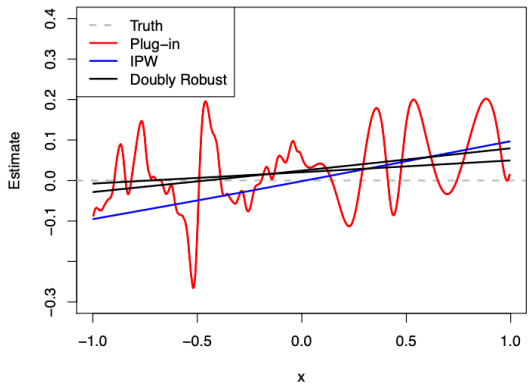
- φ is the uncentred **efficient influence function** of the ATE: AIPW, pointwise
- Its conditional mean is the CATE: $\mathrm{E}[\varphi \mid X] = \tau(X)$. Smoothing φ over X is a regression with the right target.
- Plugging in $\hat{e}, \hat{\mu}_a$ adds bias only through the **product of their errors**: doubly robust at every x.



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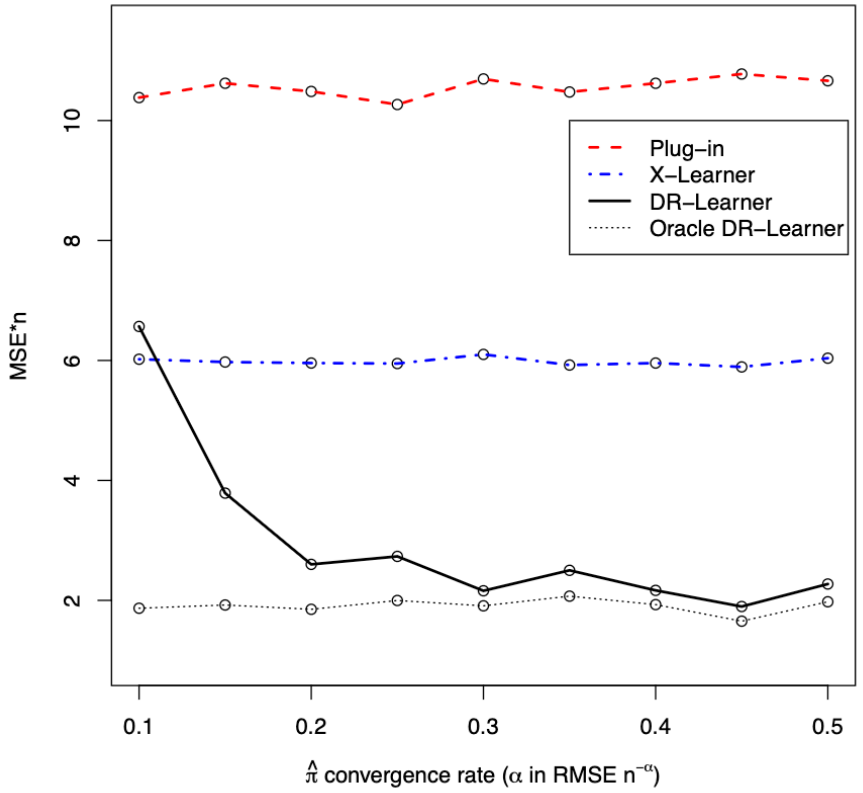
Kennedy's simulation: rough μ_a, flat τ. Only the plug-in inherits the roughness.

- Error = *oracle* error of regressing the true φ on X, plus $\hat{b}(x)^2$
- Harder than the ATE by a factor growing in $\dim X$, shrinking in τ's smoothness

$$\hat{b}(x) = \mathrm{E}[\hat{\varphi} - \varphi \mid X = x] = (e - \hat{e}) \left\{ \frac{\mu_1 - \hat{\mu}_1}{\hat{e}} + \frac{\mu_0 - \hat{\mu}_0}{1 - \hat{e}} \right\}$$

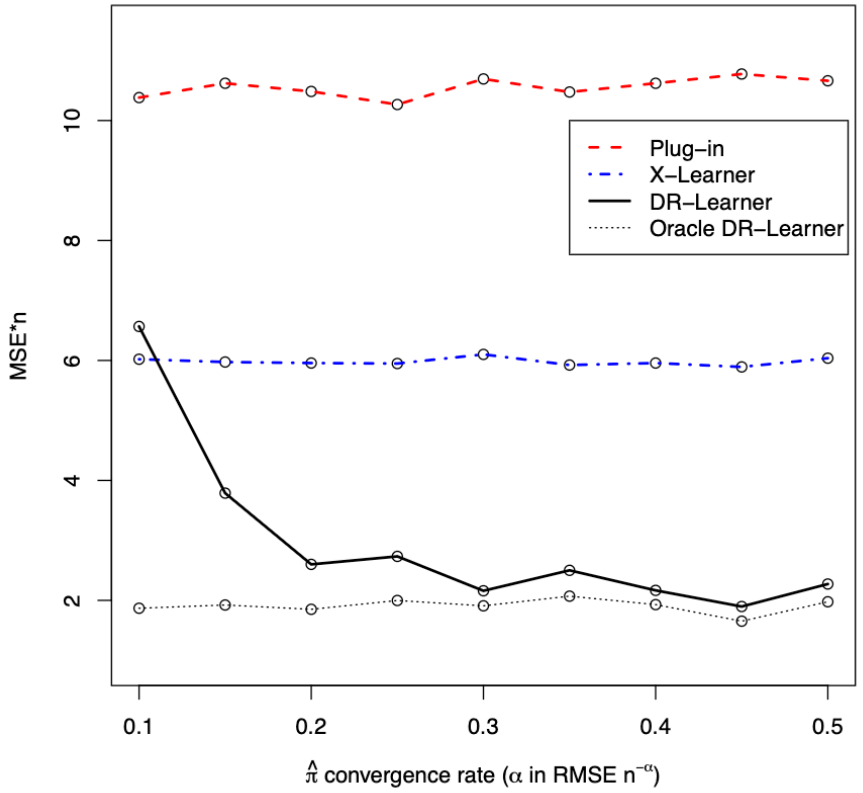
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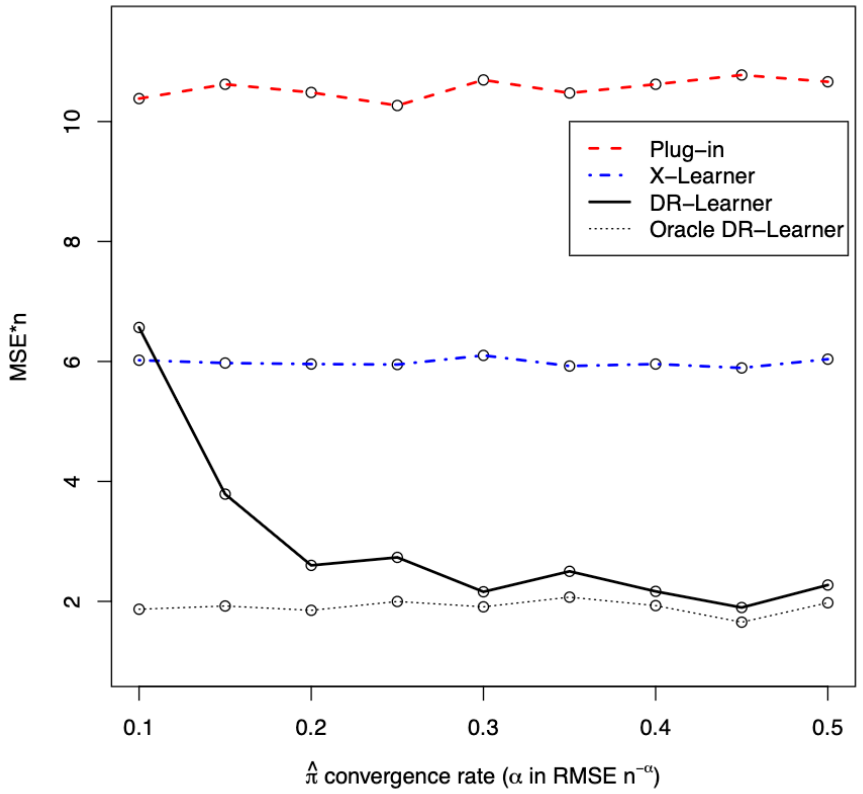
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- We're using our outcome models as a **control variate** to reduce variance
  - For a stratified second-stage model, *this is just AIPW*
- **In practice:** pseudo-outcomes are noisy (they divide by $\hat{e}(1-\hat{e})$), so clip $\hat{e}$ and regularise the second stage. In an RCT with known e, $\hat{b}(x) = 0$ however bad the $\hat{\mu}_a$ are.



References

Arbour, David, Drew Dimmery, and Anup Rao. 2021. "Efficient Balanced Treatment Assignments for Experimentation." In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics*, edited by Arindam Banerjee and Kenji Fukumizu, 130:3070–78. Proceedings of Machine Learning Research. PMLR. https://proceedings.mlr.press/v130/arbour21a.html.

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